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Another representation of the p -adic integers: In addition to computable sequences we saw yesterday, we may express p -adic integers as "infinite base p numbers." That is, we could write $a \in \mathbb{Z}_p$ as $a = a_0 + a_1 p + a_2 p^2 + \dots$ where $a_i \in \{0, 1, \dots, p-1\}$.

Given this alternative representation, one could ask how we translate between representations.

Translations: Series $\rightarrow$ sequence Given the series $a = a_0 + a_1 p + a_2 p^2 + \dots = \sum_{k=0}^{\infty} a_k p^k$, we simply build the sequence by defining each term therein as the n th partial sum reduced modulo p^n . So $x_n = \sum_{k=0}^{n-1} a_k p^k \bmod p^n$ builds $x = (x_1, x_2, x_3, \dots)$.

Sequence $\rightarrow$ series Given the sequence $x = (x_1, x_2, x_3, \dots)$ where each $x_k \in \mathbb{Z}/p^k \mathbb{Z}$ and $x_k \equiv x_{k-1} \bmod p^{k-1}$, then we want $a_0 = x_1 \bmod p$. From here we can recursively determine a_k by $a_k = \frac{x_{k+1} - x_k}{p^k} \bmod p$.

Example: Translate $(3, 13, 38, \dots)$ to a power series then back in $\mathbb{Z}_5$.

Here, $x_1 = 3 \bmod 5$, $x_2 = 13 \bmod 25$, $x_3 = 38 \bmod 125$. Thus $a_0 = 3 \bmod 5$, $a_1 = \frac{13-3}{5} = 2$, $a_2 = \frac{38-13}{25} = 1$,

and so $a = 3 + 2 \cdot 5 + 1 \cdot 5^2 + \dots$. Then going back, we have $x_1 = \sum_{k=0}^0 a_k p^k = a_0 = 3$, $x_2 = \sum_{k=0}^1 a_k p^k = a_0 + a_1 p = 3 + 2 \cdot 5 = 13$, and $x_3 = \sum_{k=0}^2 a_k p^k = a_0 + a_1 p + a_2 p^2 = 3 + 2 \cdot 5 + 1 \cdot 5^2 = 38$.

Recall that operations on sequences in $\mathbb{Z}_p$ were pointwise. Operations on power series in $\mathbb{Z}_p$ are not as straightforward and function more like operations with typical base b numbers where we must "carry" digits.

Example: Consider the 3-adic integers $s = 1 + 2 \cdot 3 + 0 \cdot 3^2 + 1 \cdot 3^3 + 2 \cdot 3^4 + \dots = \dots 21021_3$ and $t = 2 + 2 \cdot 3 + 1 \cdot 3^2 + 0 \cdot 3^3 + 0 \cdot 3^4 + \dots = \dots 00122_3$. What is $s \cdot t$?

$$s \cdot t = 2 \cdot 3^0 + (2+4) \cdot 3^1 + (0+1+4) \cdot 3^2 + (2+0+0+2) \cdot 3^3 + (4+0+0+2+0) \cdot 3^4 + \dots$$

$$= 2 \cdot 3^0 + 6 \cdot 3^1 + 5 \cdot 3^2 + 4 \cdot 3^3 + 6 \cdot 3^4 + \dots$$

$$= 2 \cdot 3^0 + 2 \cdot 3^1 + 6 \cdot 3^2 + 4 \cdot 3^3 + 6 \cdot 3^4 + \dots$$

$$= 2 \cdot 3^0 + 2 \cdot 3^1 + 2 \cdot 3^2 + 5 \cdot 3^3 + 6 \cdot 3^4 + \dots$$

$$= 2 \cdot 3^0 + 2 \cdot 3^1 + 2 \cdot 3^2 +$$

$$= 2 \cdot 3^0 + 2 \cdot 3^2 + (2+3) \cdot 3^2 + (1+3) \cdot 3^3 + 2 \cdot 3^5 + \dots$$

$$= 2 \cdot 3^0 + 0 \cdot 3^1 + (1+3) \cdot 3^2 + 3^3 + (1+3) \cdot 3^3 + 2 \cdot 3^5 + \dots$$

$$= 2 \cdot 3^0 + 0 \cdot 3^1 + 1 \cdot 3^2 + 1 \cdot 3^3 + 2 \cdot 3^4 + 2 \cdot 3^5 + \dots$$

* But then recall $a_i \in \{0, 1, \dots, p-1\}$, so we must "carry" digits *

* Note since we didn't include data up to 3^5 , its coefficient may not be 2 and it should be absorbed into the ~~ellipsis~~ ellipsis *

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The set $\mathbb{Q}_p$: We define $\mathbb{Q}_p$ to be the ~~set~~^{ring} of Laurent Series in p with finite negative portion. That is, every ~~finite~~ nonzero element $x \in \mathbb{Q}_p$ expands uniquely as $x = p^{-m}a_{-m} + \dots + p^{-1}a_{-1} + a_0 + pa_1 + p^2a_2 + \dots = \sum_{k=-m}^{\infty} a_k p^k$, where $-m$ is some finite integer. We still have $a_k \in \{0, 1, \dots, p-1\}$ and the leading coefficient $a_{-m} \neq 0$.

Connection to $\mathbb{Z}_p$:

Every nonzero element $x \in \mathbb{Q}_p$ can be written uniquely as $x = p^k \cdot u$ where $v_p(x) = k$ and $u \in \mathbb{Z}_p^\times$. Moreover, it follows that $\|x\|_p = \|p^k u\|_p = \|p^k\|_p = p^{-v_p(p^k)} = p^{-k}$.

Note: $\mathbb{Q}_p$ can be thought of as $\mathbb{Z}_p[\frac{1}{p}]$.

Theorem: $\mathbb{Q}_p$ is a field and the smallest such one containing $\mathbb{Z}_p$.

p-adic Valuation: We extend the notion of $v_p(a)$ to $\mathbb{Q}_p$ and $\mathbb{Z}_p$ by making it be the first index of $a \in \mathbb{Q}_p$ or $\mathbb{Z}_p$ for which $a_m \neq 0$. That is, if $a = a_n p^n + a_{n+1} p^{n+1} + \dots$ where n is possibly less than or equal to zero, then $v_p(a) = n$.

Ultra metric property: $\|x+y\|_p \leq \max\{\|x\|_p, \|y\|_p\} \leq \|x\|_p + \|y\|_p$.

Proof: Suppose $x, y \in \mathbb{Z}_p$. Then notice that $\max\{\|x\|_p, \|y\|_p\} = \max\{p^{-v_p(x)}, p^{-v_p(y)}\}$.

Since we have p to a negative exponent, the maximum will be the smallest exponent. That is, $\max\{p^{-v_p(x)}, p^{-v_p(y)}\} = p^{-\min\{v_p(x), v_p(y)\}}$, and since $\|x+y\|_p = p^{-v_p(x+y)}$, if we can show $-v_p(x+y) \leq -\min\{v_p(x), v_p(y)\}$, then we're done. Let $v_p(x) = a$ and $v_p(y) = b$ and suppose wlog $a \leq b$. Then we can write $x = p^a \cdot u$ and $y = p^b \cdot v$ where $p \nmid u, v$. Then since $a \leq b$, $y = p^a(p^{b-a} \cdot v)$ and so $x+y = p^a(u + p^{b-a} \cdot v)$ implying that $p^a \mid x+y$ and therefore $v_p(x+y) \geq a = \min\{v_p(x), v_p(y)\}$. Lastly, $\max(s, t) \leq s+t$ because wlog if $0 \leq s \leq t$, then $\max(s, t) = s \leq s+t$. $\square$

Theorem: $\mathbb{Q}_p$ is a complete space wrt the p-adic metric $\|\cdot\|_p = p^{-v_p(\cdot)}$.

↳ Homework

Note: Much of the phenomena in $\mathbb{F}_p$ lifts to $\mathbb{Z}_p$. For example, $a \in \mathbb{Z}_p$ is a unit iff $a \bmod p$ is a unit in $\mathbb{F}_p$. This leads us to the following result.

Hensel's Lemma: If $f(x) \in \mathbb{Z}_p[x]$ and $a \in \mathbb{Z}_p$ satisfies $f(a) \equiv 0 \bmod p$ and $f'(a) \not\equiv 0 \bmod p$, then there is a unique $\alpha \in \mathbb{Z}_p$ s.t. $f(\alpha) = 0$ in $\mathbb{Z}_p$ and $\alpha \equiv a \bmod p$.

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p-adic numbers: We've been studying integers modulo p , but this causes us to lose a lot of information. Thus in order to preserve more data, we look to another consideration of integers.

Theorem: We have $m=n$ iff $m \equiv n \pmod{p^k}$ for all k .

p-adic integers: For each prime p , we let $\mathbb{Z}_p$ be the ring of p-adic integers. The elements of this ring are sequences $x = (x_1 \pmod{p}, x_2 \pmod{p^2}, \dots)$ which satisfy the compatibility relations $x_i \equiv x_j \pmod{p^i}$ for $i < j$.

Example: An element and non-element of $\mathbb{Z}_7$.

The infinite sequence $(1, 15, 64, \dots)$ lies in $\mathbb{Z}_7$ because $1 \in \mathbb{Z}/7\mathbb{Z}$, $15 \in \mathbb{Z}/7^2\mathbb{Z}$, $64 \in \mathbb{Z}/7^3\mathbb{Z}$, $1 \equiv 15 \pmod{7}$, and $64 \equiv 15 \pmod{7^2}$.

The infinite sequence $(1, 2, 51, \dots) \notin \mathbb{Z}_7$ since $1 \in \mathbb{Z}/7\mathbb{Z}$, $2 \in \mathbb{Z}/7^2\mathbb{Z}$, $51 \in \mathbb{Z}/7^3\mathbb{Z}$, but $1 \not\equiv 2 \pmod{7}$.

Question: Something we could ask ourselves is how many possibilities are there for each step of building sequence which lie in $\mathbb{Z}_p$?

Answer: We build these sequences element by element. For the first entry, x_1 , there are $|\{0, 1, \dots, p-1\}| = p$ choices. Then for a given x_1 , recall we require $x_1 \equiv x_2 \pmod{p}$ which is $x_2 = 7k + x_1$ for which there are again p choices for k . Thus at each step, there are p choices for each entry.

Operations: The multiplication and addition of elements in $\mathbb{Z}_p$ is pointwise and functions as typical multiplication and addition of elements in $\mathbb{Z}/p\mathbb{Z}$.

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Example: In $\mathbb{Z}_3$, consider $x = (1, 4, 13, \dots)$ and $y = (2, 8, 11, \dots)$.
Then $x+y = (3, 12, 24, \dots) = (0, 3, 24, \dots)$ and
 $x \cdot y = (2, 32, 143, \dots) = (2, 5, 8, \dots)$.

Identities: The additive identity of $\mathbb{Z}_p$ is $(0, 0, 0, \dots)$ and the multiplicative identity is $(1, 1, 1, \dots)$.

Injection: The map $\mathbb{Z} \rightarrow \mathbb{Z}_p$ given by $n \mapsto (n \bmod p, n \bmod p^2, \dots)$ forms an injection.

Theorem: If $a \cdot b = 0$ in $\mathbb{Z}_p$, then either a or b was zero. That is, $\mathbb{Z}_p$ is an integral domain.

Note: $\mathbb{Z}_p$ has the stronger property that it is a PID.

Inverse: Any p -adic integer $x = (x_1, x_2, x_3, \dots)$ is invertible in $\mathbb{Z}_p$ as long as $x_1 \not\equiv 0 \pmod p$.

Units: These are of course the invertible elements. Thus an element is a unit iff $x_1 \not\equiv 0 \pmod p$, ~~not~~

Example: Find the inverse of $(3, 3, 3, \dots)$ in $\mathbb{Z}_5$.

Since $x_1 = 3 \not\equiv 0 \pmod 5$, this is a unit. We want the inverse $y = (y_1, y_2, y_3, \dots)$ which must satisfy $x_1 y_1 \equiv 1 \pmod 5$. ~~and~~ That is, $3 \cdot y_1 \equiv 1 \pmod 5$. By inspection, 2 works. Then we require $y_1 \equiv y_2 \pmod 5$ and $x_2 y_2 \equiv 1 \pmod{25}$. There are 5 possibilities for $y_2 \equiv 2 \pmod 5$, $\{2, 7, 12, 17, 22\}$ of which, $3 \cdot 17 \equiv 1 \pmod{25}$.

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Example cont:

We then require $y_3 \equiv 17 \pmod{25}$ and $3 \cdot y_3 \equiv 1 \pmod{125}$. For $y_3 \equiv 17 \pmod{25}$, there are possibilities $\{17, 42, 67, 92, 117\}$, of which $3 \cdot 42 \equiv 1 \pmod{25}$. Therefore, the inverse sequence $y = (2, 17, 42, \dots)$ represents the 5-adic integer $\frac{1}{3}$.

Example: Construct the 3-adic integer $\frac{1}{7}$.

Let $x = (7, 7, 7, \dots)$ in $\mathbb{Z}_3$. Then we want the inverse $y = (y_1, y_2, y_3, \dots)$ such that $y_1 \cdot 7 \equiv 1 \pmod{3}$. Upon inspection out of the choices $\{0, 1, 2\}$ $y_1 = 1$ works. Then we require $y_2 \equiv 1 \pmod{3}$ and $7 \cdot y_2 \equiv 1 \pmod{3^2}$. We have choices $\{1, 4, 7\}$ of which $y_2 = 4$ works. Lastly, we require $y_3 \equiv 4 \pmod{3^2}$ and $7 \cdot y_3 \equiv 1 \pmod{3^3}$. We have choices $\{4, 13, 22\}$ of which $y_3 = 4$ works. Thus the 3-adic integer $\frac{1}{7}$ is $y = (1, 4, 4, \dots)$.

↓
Valuation & Completeness: We consider the p-adic absolute value on $\mathbb{Q}$,
 $\|x\|_p := p^{-v_p(x)}$

Properties: (1) Positive definite (2) Multiplicative (3) Satisfies triangle inequality.

Proof:

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Example: Determine the squareness of 5 modulo 11.

Here we have $a=5$ and $p=11$, so $\frac{p-1}{2}=5$. First, the squares modulo 11 are: $0^2 \equiv 0, 1^2 \equiv 1, 2^2 \equiv 4, 3^2 \equiv 9, 4^2 \equiv 5, 5^2 \equiv 3 \pmod{11}$ and so we clearly see, yes, indeed 5 is a square modulo 11 and in fact $5 \equiv 4^2 \pmod{11}$. However, we would of course like to use Gauss' lemma. We have $I = \{1, 2, 3, 4, 5\}$ and $a \cdot I = \{5, 10, 15, 20, 25\} \equiv \{5, 10, 4, 9, 3\}$. Therefore $n=2$ which is even and so 5 is a square by Gauss' lemma.

Corollary: $2 \equiv x^2 \pmod{p}$ iff $p \equiv \pm 1 \pmod{8}$. *where p is odd*

Proof: First, since $\mathbb{Z}/8\mathbb{Z} = \{0, 1, 2, 3, 4, 5, 6, 7\}$ and p is odd, the only possible cases for p are 1, 3, 5, 7 modulo 8, so there are only 4 possible cases. First though, here $I = \{1, 2, \dots, \frac{p-1}{2}\}$ and $2I = \{2, 4, \dots, p-1\}$. Then, an element $2k \in 2I$ has $1 \leq k \leq \frac{p-1}{2}$ and is not in I after the restriction of $2I$ modulo p iff $2k > \frac{p}{2}$ or $k > \frac{p}{4}$. We therefore have n will be the number of elements for which $\frac{p}{4} < k \leq \frac{p-1}{2}$.

$p \equiv 1 \pmod{8}$: If $p \equiv 1 \pmod{8}$, then $p = 8q + 1$ and so $\frac{8q+1}{4} < k \leq \frac{8q}{2}$ implies that $2q + \frac{1}{4} < k \leq 4q$ and so $k \in \{2q+1, 2q+2, \dots, 4q\}$ which has length $4q - (2q+1) + 1 = 2q$ and so n is even which happens iff 2 is a square.

$p \equiv 3 \pmod{8}$: If $p \equiv 3 \pmod{8}$, then $p = 8q + 3$ and so $\frac{8q+3}{4} < k \leq \frac{8q+3-1}{2}$ implies that $2q + \frac{3}{4} < k \leq 4q + 1$. Thus $k \in \{2q+1, 2q+2, \dots, 4q+1\}$ which has length $4q+1 - (2q+1) + 1 = 2q+1$ which is odd. Therefore this happens iff 2 is not a square.

$p \equiv 5 \pmod{8}$: If $p \equiv 5 \pmod{8}$, then $p = 8q + 5$ and so $\frac{8q+5}{4} < k \leq \frac{8q+5-1}{2}$ implies that $2q + \frac{5}{4} < k \leq 4q + 2$. Thus $k \in \{2q+2, 2q+3, \dots, 4q+2\}$ which has length $4q+2 - (2q+2) + 1 = 2q+1$. Thus n is odd which happens iff 2 is not a square.

$p \equiv 7 \pmod{8}$: If $p \equiv 7 \pmod{8}$, then $p = 8q + 7$ and so $\frac{8q+7}{4} < k \leq \frac{8q+7-1}{2}$ implies that $2q + \frac{7}{4} < k \leq 4q + 3$. Thus $k \in \{2q+2, 2q+3, \dots, 4q+3\}$ which has length $4q+3 - (2q+2) + 1 = 2q+2 = 2(q+2)$. Therefore n is even which happens iff 2 is a square. Lastly all we must do is notice $p \equiv 7 \equiv -1 \pmod{8}$.

□

Discriminant: Given a polynomial $p(x) = ax^2 + bx + c$ the discriminant of $p(x)$ is the quantity $b^2 - 4ac$. This is related to the roots of $p(x)$ via the quadratic formula.

Example: (field discriminant)

Consider the minimal monic polynomial of α $p(x) = \begin{cases} x^2 - d, & d \not\equiv 1 \pmod{4} \\ x^2 - x + \frac{1-d}{4}, & d \equiv 1 \pmod{4} \end{cases}$. Then if $K = \mathbb{Q}[\sqrt{d}]$, we call Δ_K the field discriminant. In particular, if $d \not\equiv 1 \pmod{4}$, then $p(x) = x^2 - d$ and so $\Delta_K = 0^2 - 4(1)(-d) = 4d$. Notice that if $q = 2$, then when $d \not\equiv 1 \pmod{4}$, q should ramify. Further notice that $2 \mid \Delta_K$ because $\Delta_K = 4d = 2K$ precisely when $K = 2d$. Then if $d \equiv 1 \pmod{4}$, $p(x) = x^2 - x + \frac{1-d}{4}$, for which $\Delta_K = (-1)^2 - 4(1)(\frac{1-d}{4}) = 1 - (1-d) = d$. As a result of this, we may reduce our cases for ramification of q dependent on the congruence of d modulo 4, to simply requiring $q \mid \Delta_K$ for q to ramify in $\mathcal{O}_K$.

Quadratic field: We have previously proven that $\mathbb{Q}[\sqrt{d}]$ is a field within the larger field $\mathbb{C}$. We give $\mathbb{Q}[\sqrt{d}]$ the additional name "quadratic field" to acknowledge that $\sqrt{d}$ is the root of a polynomial of degree two.

Dedekind - Kummer Theorem: Suppose $\mathcal{O}_K = \mathbb{Z}[x]$ and $q \in \mathbb{Z}$ is a prime. Further, modulo q , $f(x)$ factors as $f(x) \equiv f_1(x)^{e_1} f_2(x)^{e_2} \dots f_g(x)^{e_g} \pmod{q}$ such that each $f_i(x)$ is irreducible and monic in $\mathbb{F}_q[x]$. Then the ideal $q\mathcal{O}_K$ factors into prime ideals $\mathfrak{p}_i^{e_i}$ as $q\mathcal{O}_K = \mathfrak{p}_1^{e_1} \mathfrak{p}_2^{e_2} \dots \mathfrak{p}_g^{e_g}$.

Euler's Criterion: Given an odd prime p , and $a \in \mathbb{Z}$, then $a \equiv x^2 \pmod{p}$ iff $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$.

Proof: Suppose $a \equiv x^2 \pmod{p}$. Then this implies $a^{\frac{p-1}{2}} \equiv (x^2)^{\frac{p-1}{2}} \equiv x^{p-1} \pmod{p}$. By Fermat's little theorem, $x^{p-1} \equiv 1 \pmod{p}$ and so $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$.

Conversely, Suppose $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$. Suppose that g is a primitive element of $\mathbb{Z}/p\mathbb{Z}$. Then this implies for some k , $g^k \equiv a \pmod{p}$. We therefore have $a^{\frac{p-1}{2}} \equiv g^{k\frac{p-1}{2}} \equiv 1 \pmod{p}$. Since g is primitive, it has order $p-1$ which implies $p-1 \mid k\frac{p-1}{2}$ or that $\frac{k}{2} \in \mathbb{Z}$. Therefore, k must be even, say $k = 2m$. we thus have $g^k \equiv g^{2m} \equiv (g^m)^2 \equiv a \pmod{p}$. $\square$

Gauss' Lemma: Let p be an odd prime and $a \in \mathbb{Z}$ s.t. $a \not\equiv 0 \pmod{p}$. Then consider the remainders $I = \{1, 2, 3, \dots, \frac{p-1}{2}\}$ and the classes $a \cdot I = \{a, 2a, 3a, \dots, (\frac{p-1}{2})a\}$. Reduce every number in $a \cdot I$ modulo p and let n be the number of elements in $a \cdot I$ but not in I . Then $a \equiv x^2 \pmod{p}$ iff n is even. In other words, let n be the number of elements $ak \in aI$ such that $ak > \frac{p}{2}$.

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Theorem: Let $d \in \mathbb{Z}$ be square free and $d \neq 0, 1$. Also let $P(x)$ be the minimal monic polynomial of α . Then exactly one of the following can happen:

(1) $P(x) \bmod q$ has a repeated factor iff

(1.1) $d \not\equiv 1 \pmod{4}$: $q=2$ or $q|d$

(1.2) $d \equiv 1 \pmod{4}$: $q \nmid d$

(2) $P(x) \bmod q$ splits into distinct linear factors iff $d \equiv x^2 \pmod{q}$ for some $x \neq 0$

(3) $P(x) \bmod q$ is irreducible iff $d \not\equiv x^2 \pmod{q}$ for any x .

Proof:

Recall: For a prime $q \in \mathbb{Z}$,

- (1) q splits iff $p(x) \bmod q$ factors into distinct linear factors, so $\mathcal{O}_q/\mathcal{O} \cong \mathbb{F}_q \times \mathbb{F}_q$.
- (2) q is inert iff $p(x) \bmod q$ is irreducible, so $\mathcal{O}_q/\mathcal{O} \cong \mathbb{F}_{q^2}$.
- (3) q ramifies iff $p(x) \bmod q$ has repeated factors, so $\mathcal{O}_q/\mathcal{O} \cong \mathbb{F}_q[x]/\langle x^2 \rangle$ and has nilpotents.

Prime ideal: We say an ideal I of R is prime when $ab \in I$ implies $a \in I$ or $b \in I$.

Example: Show that $\langle 2, x^2+x+1 \rangle \subset \mathbb{Z}[x]$ is a prime ideal.

First, recall P is a prime ideal of the ring R iff R/P is an integral domain. We thus would like to show $\mathbb{Z}[x]/\langle 2, x^2+x+1 \rangle$ is an integral domain. Well, recall we had that $\mathbb{Z}[x]/\langle q, p(x) \rangle \cong \mathbb{F}_q[x]/\langle p(x) \rangle$ and so $\mathbb{Z}[x]/\langle 2, x^2+x+1 \rangle \cong \mathbb{F}_2[x]/\langle x^2+x+1 \rangle$. Then notice that x^2+x+1 is irreducible in $\mathbb{F}_2[x]$ and so this is a field of 4 elements. Since fields are integral domains, we are done. $\square$

Note: Prime ideals give a generalization of how primes factor, even when the ring under consideration is not a UFD. For example, in $\mathbb{Z}[\sqrt{-5}]$.

Operations on ideals: Let I, J be ideals of a ring R . Then $I+J = \{x+y \mid x \in I, y \in J\}$ and $I \cdot J = \langle \{x \cdot y \mid x \in I, y \in J\} \rangle$ ideal generated by the set of products $\{x \cdot y \mid x \in I, y \in J\}$.

Example: consider $\langle 2 \rangle \subset \mathbb{Z}[\sqrt{-5}]$.

We consider $\langle 2, 1+\sqrt{-5} \rangle$ which is not a prime ideal, but we claim $\langle 2, 1+\sqrt{-5} \rangle^2 = \langle 2 \rangle$ which shows that 2 ramifies in $\mathbb{Z}[\sqrt{-5}]$. To see this we vts $\langle 2 \rangle \subset \langle 2, 1+\sqrt{-5} \rangle^2$ and $\langle 2, 1+\sqrt{-5} \rangle^2 \subset \langle 2 \rangle$. First, we consider what $\langle 2, 1+\sqrt{-5} \rangle^2$ is. Well, $\langle 2, 1+\sqrt{-5} \rangle^2 = \langle 2^2, 2(1+\sqrt{-5}), (1+\sqrt{-5})^2 \rangle = \langle 4, 2+2\sqrt{-5}, -4+2\sqrt{-5} \rangle$. Then, notice every element in $\langle 2, 1+\sqrt{-5} \rangle^2$ is a multiple of 2, so we have $\langle 2, 1+\sqrt{-5} \rangle^2 \subset \langle 2 \rangle$. Then to see $\langle 2 \rangle \subset \langle 2, 1+\sqrt{-5} \rangle^2$, we'd like to show $2 \in \langle 2, 1+\sqrt{-5} \rangle^2$. Recall ideals are closed under addition and so $\langle 2, 1+\sqrt{-5} \rangle^2 \ni (2+2\sqrt{-5}) + (-4+2\sqrt{-5}) = 2$. Therefore we have $\langle 2, 1+\sqrt{-5} \rangle^2 = \langle 2 \rangle$ and so 2 ramifies in $\mathbb{Z}[\sqrt{-5}]$. $\square$

Theorem: Let $\mathcal{O}$ be the ring of algebraic integers. Every ideal of $\mathcal{O}$ factors uniquely into prime ideals.

Generalized CRT: Let R be a ring, I and J ideals, and $I+J=R$. Then $I \cdot J = I \cap J$ and $R/I \cdot J \cong R/I \times R/J$.

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Let $d \in \mathbb{Z}$ be square free, $K = \mathbb{Q}[\sqrt{d}]$, $\alpha = \begin{cases} \sqrt{d}, & d \not\equiv 1 \pmod{4} \\ \frac{1+\sqrt{d}}{2}, & d \equiv 1 \pmod{4} \end{cases}$,
and $\mathcal{O} = \mathbb{Z}[\alpha]$.

Yesterday, we showed $\mathcal{O} \subset K \cap \overline{\mathbb{Z}}$. We would now like to show $K \cap \overline{\mathbb{Z}} \subset \mathcal{O}$ which in turn shows $\mathcal{O} = K \cap \overline{\mathbb{Z}}$.

Lemma: Let $\beta \in \overline{\mathbb{Q}}$. Then let $p(x) \in \mathbb{Q}[x]$ be the monic polynomial of smallest degree with $p(\beta) = 0$. Then $\beta \in \overline{\mathbb{Z}}$ iff $p(x) \in \mathbb{Z}[x]$.

Gauss' Lemma: Suppose $A(x) \in \mathbb{Z}[x]$ s.t. $\gcd(\text{coefficients of } A) = 1$. Then $A(x)$ is irreducible in $\mathbb{Z}[x]$ iff $A(x)$ is irreducible in $\mathbb{Q}[x]$.

Theorem: $K \cap \overline{\mathbb{Z}} \subset \mathcal{O}$.

Pseudo proof: Let $x + y\sqrt{d} \in K \cap \overline{\mathbb{Z}}$. Then $(T - (x + y\sqrt{d}))(T - (x - y\sqrt{d})) \in \mathbb{Z}[T]$ and so since $(T - (x + y\sqrt{d}))(T - (x - y\sqrt{d})) = T^2 - 2Tx + (x^2 - dy^2)$, we have $2x \in \mathbb{Z}$ and $x^2 - dy^2 \in \mathbb{Z}$. Lastly, $2x \in \mathbb{Z}$ implies either $x \in \mathbb{Z}$ or $x = x'/2$ with $x' \in \mathbb{Z}$ and odd. We then have two cases to cover since there are two possibilities for x . We first suppose $d \not\equiv 1 \pmod{4}$ and then NTS $x \in \mathbb{Z}$ and $y \in \mathbb{Z}$. We then suppose $d \equiv 1 \pmod{4}$ and NTS either $x \in \mathbb{Z}$ and $y \in \mathbb{Z}$ or $x = x'/2$ with $x' \in \mathbb{Z}$ odd and $y = y'/2$ with $y' \in \mathbb{Z}$ odd.

Note: All of these ideas are very important to how primes split in rings.

Corollary: $\mathcal{O} \cong \mathbb{Z}[x]/p(x)$ where $p(x)$ is the monic minimal polynomial of α with integer coefficients, so $p(x) = \begin{cases} x^2 - d, & d \not\equiv 1 \pmod{4} \\ x^2 - x + \frac{1-d}{4}, & d \equiv 1 \pmod{4} \end{cases}$.

Chain: We claim $\mathcal{O}/q\mathcal{O} \cong \mathbb{Z}[x]/p(x) / q \cdot (\mathbb{Z}[x]/p(x)) \cong \mathbb{Z}[x] / (q, p(x)) \cong \mathbb{F}_q[x] / (\overline{p(x)})$

where q is some prime in $\mathbb{Z}$.

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Idea: Given $\mathcal{O}/\mathfrak{q} \cdot \mathcal{O} \cong \mathbb{F}_q[X]/\langle \bar{p}(X) \rangle$ where $\bar{p}(X) \in \mathbb{F}_q[X]$ is the reduction of the minimal polynomial $p(X)$ modulo $\mathfrak{q}$. As we saw, there are two cases: (1) If $d \not\equiv 1 \pmod{4}$, $p(X) = X^2 - d$. (2) If $d \equiv 1 \pmod{4}$, $p(X) = X^2 - X + \frac{1-d}{4}$.

We have exactly 3 possibilities for the factorization of $\bar{p}(X)$ corresponding to the behavior of $\mathfrak{q}$ in $\mathcal{O}$. They are:

Behavior of $\mathfrak{q}$	Factorization of $\bar{p}(X)$ in $\mathbb{F}_q[X]$	Isomorphism class of $\mathcal{O}/\mathfrak{q}\mathcal{O}$
Inert	Irreducible quadratic polynomial	$\mathbb{F}_{q^2}$: a finite field with q^2 elements
Split	Two distinct factors: $(X-a)(X-b)$	$\mathbb{F}_q \times \mathbb{F}_q$: Product ring of two fields
Ramified	Repeated linear factor: $(X-a)^2$	$\mathbb{F}_q[X]/\langle (X-a)^2 \rangle$ w/ $\bar{p}(a) = 0$

Example: Consider $\mathbb{Z}[i]/3\mathbb{Z}[i]$, $\mathbb{Z}[i]/5\mathbb{Z}[i]$, $\mathbb{Z}[i]/2\mathbb{Z}[i]$.

Here $d = -1$ and so $d \not\equiv 1 \pmod{4}$. Therefore $p(X) = X^2 - (-1) = X^2 + 1$.

Then for $\mathbb{Z}[i]/3\mathbb{Z}[i]$, $q = 3$ and since $p(X) \neq 0$ in $\mathbb{F}_3[X]$, $p(X)$ is irreducible. By our above chart, this implies 3 is inert and $\mathbb{Z}[i]/3\mathbb{Z}[i] \cong \mathbb{F}_9$. For $\mathbb{Z}[i]/5\mathbb{Z}[i]$, $q = 5$ and we consider $p(X)$ in $\mathbb{F}_5[X]$. We have $p(0) = 1$, $p(1) = 2$, $p(2) = 5 \equiv 0 \pmod{5}$, $p(3) = 10 \equiv 0 \pmod{5}$, $p(4) = 17 \equiv 2 \pmod{5}$. So $\bar{p}(X) = (X-2)(X-3)$ whence 5 splits in $\mathbb{Z}[i]$.

We have $\mathbb{Z}[i]/5\mathbb{Z}[i] \cong \mathbb{F}_5 \times \mathbb{F}_5$. Lastly, for $\mathbb{Z}[i]/2\mathbb{Z}[i]$ we have $q = 2$. Then $\bar{p}(0) = 1$, $\bar{p}(1) = 0$ and so $\bar{p}(X) = (X-1)^2$ in which $\mathbb{Z}[i]/2\mathbb{Z}[i] \cong \mathbb{F}_2[X]/\langle (X-1)^2 \rangle$ and implying 2 ramifies in $\mathbb{Z}[i]$.

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Question: Can we generalize the idea $p \equiv 1 \pmod{4}$ iff there exists some $x \in \mathbb{Z}$ s.t. $-1 \equiv x^2 \pmod{p}$ to a congruence condition on p for when there exists some $y \in \mathbb{Z}$ s.t. $d \equiv y^2 \pmod{p}$? That is, can we generalize the ideas in $\mathbb{Z}[i]$ to $\mathbb{Z}[\sqrt{d}]$?

It turns out that we indeed can and the idea of this generalization gives rise to what is known as Quadratic Reciprocity, we touch on this more later.

Definition: We say $\alpha \in \mathbb{C}$ is an algebraic number when α is a root of a polynomial with rational coefficients. Then an algebraic integer is an algebraic number that is the root of a monic polynomial. We then write $\bar{\mathbb{Z}}$ to denote the ring of algebraic integers and $\bar{\mathbb{Q}}$ to denote the field of algebraic numbers.

Lemma: $\mathbb{Q}[\sqrt{d}]$ is a subfield of $\mathbb{C}$ for $d \in \mathbb{Z}$.

Proof: This claims that $\mathbb{Q}[\sqrt{d}]$ is a subset of $\mathbb{C}$ and that it is a field. It is immediately obvious that $\mathbb{Q}[\sqrt{d}]$ is a subset of $\mathbb{C}$. Then the claim that it is a field can be broken into: it is a ring and all nonzero elements have an inverse. To show it is a ring, we must show it is closed under addition and multiplication and also contains 1, ~~and~~ 0, and additive inverses. The existence of 1, 0, and additive inverses is clear and so is closure under addition, what may be less clear is multiplication. We have $(a+b\sqrt{d})(c+e\sqrt{d}) = (ac+bed) + (ae+bc)\sqrt{d} \in \mathbb{Q}[\sqrt{d}]$, so this is a ring. To show this is a field, we must show every nonzero element has an inverse. Consider $a+b\sqrt{d} \neq 0$. We then show $\frac{1}{a+b\sqrt{d}} \in \mathbb{Q}[\sqrt{d}]$. Notice $\frac{1}{a+b\sqrt{d}} \cdot \frac{a-b\sqrt{d}}{a-b\sqrt{d}} = \frac{a-b\sqrt{d}}{a^2-b^2d}$ where $a^2-b^2d \neq 0$ and is rational. Therefore, $\frac{1}{a+b\sqrt{d}} = \frac{a}{a^2-b^2d} + \frac{(-b)\sqrt{d}}{a^2-b^2d} \in \mathbb{Q}[\sqrt{d}]$, proving $\mathbb{Q}[\sqrt{d}]$ is a field $\square$

Theorem: Let $d \in \mathbb{Z}$ be square free and $d \neq 0, 1$. Then the ring of algebraic integers in $\mathbb{Q}[\sqrt{d}]$ is:
(1) $\mathbb{Z}[\sqrt{d}]$ if $d \not\equiv 1 \pmod{4}$, (2) $\mathbb{Z}[\frac{1+\sqrt{d}}{2}]$ if $d \equiv 1 \pmod{4}$.

Proof: We first NTS $\mathbb{Z}[\sqrt{d}]$ and $\mathbb{Z}[\frac{1+\sqrt{d}}{2}]$ are rings. The case of $\mathbb{Z}[\sqrt{d}]$ should be clear from the above lemma. For $\mathbb{Z}[\frac{1+\sqrt{d}}{2}]$, ~~this is clear~~ we must show closure under addition and multiplication as well as containment of 0, 1, and additive inverses. All of these are clear except for closure of multiplication. We have $(x+y(\frac{1+\sqrt{d}}{2}))(z+w(\frac{1+\sqrt{d}}{2})) = xz + (yz+xw)(\frac{1+\sqrt{d}}{2}) + yw(\frac{1+\sqrt{d}}{2})^2$. Then $xz + (yz+xw)(\frac{1+\sqrt{d}}{2}) \in \mathbb{Z}[\frac{1+\sqrt{d}}{2}]$ so all we NTS is $yw(\frac{1+\sqrt{d}}{2})^2 \in \mathbb{Z}[\frac{1+\sqrt{d}}{2}]$. Recall $d \equiv 1 \pmod{4}$ and so $d = 4k+1$ for some $k \in \mathbb{Z}$. Then $(\frac{1+\sqrt{d}}{2})^2 = \frac{1+2\sqrt{d}+d}{4} = \frac{2+2\sqrt{d}+4k}{4} = k + \frac{1+\sqrt{d}}{2} \in \mathbb{Z}[\frac{1+\sqrt{d}}{2}]$. This shows they're both rings. We then NTS that for every $\alpha \in \mathbb{Z}[\sqrt{d}]$ we have $\alpha \in \bar{\mathbb{Z}}$ and for every $\beta \in \mathbb{Z}[\frac{1+\sqrt{d}}{2}]$, $\beta \in \bar{\mathbb{Z}}$. To see this, given $a+b\sqrt{d} \in \mathbb{Z}[\sqrt{d}]$ we claim $(x-(a+b\sqrt{d}))(x-(a-b\sqrt{d})) \in \mathbb{Z}[x]$ is monic with integer coefficients. Similarly, if $d \equiv 1 \pmod{4}$, we consider $c+d\sqrt{d} \in \mathbb{Z}[\frac{1+\sqrt{d}}{2}]$ and $(x-(c+d\sqrt{d}))(x-(c-d\sqrt{d})) \in \mathbb{Z}[x]$, claiming this is monic with integer coefficients. $\square$

→ Earlier in the course, we saw that $\mathbb{Z}[i]$ is a UFD. We may then ask, for each prime $p \in \mathbb{Z}$, how does p factor in $\mathbb{Z}[i]$?

The sum of squares theorem gave us a nice way to classify the results. Notice if $p = 2$, then p factors as $p = (1+i)(1-i)$. Then if $p \equiv 1 \pmod{4}$ we have $p = x^2 + y^2$ in which $x^2 + y^2 = (x+iy)(x-iy)$ where $x+iy$ and $x-iy$ are distinct prime factors in $\mathbb{Z}[i]$. Lastly, if $p \equiv 3 \pmod{4}$, then p is prime in $\mathbb{Z}[i]$ or is irreducible in $\mathbb{Z}[i]$.

→ This above example gives rise to a more general definition.

Definition: For $p \in \mathbb{Z}$ and a UFD R , we say p :

- (1) ramifies if p can be expressed in repeated factors (up to associateness) in R ,
- (2) splits if p is not prime and splits into distinct factors in R ,
- (3) is inert if p remains prime in R .

→ Example: Since $3 \equiv 3 \pmod{4}$, we say 3 is inert in $\mathbb{Z}[i]$. Since $5 \equiv 1 \pmod{4}$, we say 5 splits in $\mathbb{Z}[i]$. In fact, it is easy to see: $5 = (2-3i)(2+3i)$. Lastly, perhaps harder to see is that 2 ramifies in $\mathbb{Z}[i]$. This is because $2 = -i(1+i)^2$ and $-i$ is a unit in $\mathbb{Z}[i]$.

Idea: To see where this comes from, consider $x^2 + 1 \in \mathbb{Z}[x]$. We will then see the behavior of p in $\mathbb{Z}[i]$ mirrors that of $x^2 + 1 \pmod{p}$.

Ramified case: Recall, we have that $x^2 + 1 \pmod{2} = (x+1)^2 \pmod{2}$.

Split case: If $p \equiv 1 \pmod{4}$, then we know there exists some a s.t. $a^2 \equiv -1 \pmod{p}$. We then have $x^2 + 1 \equiv (x+a)(x-a) \pmod{p}$ where $a \not\equiv -a \pmod{p}$.

Inert case: If $p \equiv 3 \pmod{4}$, then -1 is not a square mod p and so $x^2 + 1$ has no roots modulo p . Since $x^2 + 1$ has degree 2, this implies it's irreducible in $\mathbb{F}_p[x]$.

→ Thus what we're really studying here is whether or not -1 is a square modulo p . This then explains why we considered primes modulo 4 in $\mathbb{Z}[i]$.

Generalizing: Suppose we wanted to generalize this concept to $\mathbb{Z}[\sqrt{d}]$ for $d \in \mathbb{Z}$.

We should consider: (1) conditions when $d \equiv x^2 \pmod{p}$

(2) If $\mathbb{Z}[\sqrt{d}]$ is even a UFD, (3) If $\mathbb{Z}[\sqrt{d}]$ is a "well behaved" ring.

→ Note: We say $\mathbb{Z}[\sqrt{d}]$ is well behaved when it is the subring of $\mathbb{Q}[\sqrt{d}]$ consisting of all algebraic integers with $\mathbb{Q}[\sqrt{d}]$.

Theorem: For all odd integers m, n s.t. $\gcd(m, n) = 1$, $(\frac{n^2-m^2}{2}, mn, \frac{n^2+m^2}{2})$ is a ~~primitive~~ PPT and distinct (m, n) give rise to distinct PPT's.

Proof: We need to show a few things: (1) $(\frac{n^2-m^2}{2}, mn, \frac{n^2+m^2}{2})$ is a Pythagorean triple, (2) $(\frac{n^2-m^2}{2}, mn, \frac{n^2+m^2}{2})$ is primitive, and (3) the pair (m, n) is unique.

(1) Notice $(\frac{n^2-m^2}{2})^2 + (mn)^2 = \frac{1}{4}(n^2-m^2)^2 + \frac{1}{4}(m^2n^2 \cdot 4) = \frac{n^4 - 2n^2m^2 + m^4 + 4m^2n^2}{4} = \frac{n^4 + 2m^2n^2 + m^4}{4} = (\frac{n^2+m^2}{2})^2$.

(2) Here we NTS $\gcd(\frac{n^2-m^2}{2}, mn) = \gcd(mn, \frac{n^2+m^2}{2}) = \gcd(\frac{n^2-m^2}{2}, \frac{n^2+m^2}{2}) = 1$.

First, $\gcd(\frac{n^2-m^2}{2}, mn) = \gcd(\frac{n^2-m^2}{2}, m) \gcd(\frac{n^2-m^2}{2}, n)$. Now suppose $g = \gcd(\frac{n^2-m^2}{2}, m)$ so that $g | \frac{n^2-m^2}{2}, m$. Then $g | \frac{n^2-m^2}{2}$ implies $\frac{n^2-m^2}{2} = gk$ or $n^2-m^2 = 2gk$ for some $k \in \mathbb{Z}$. We therefore have that $g | n^2$ and then $g | m$ implies $g | m^2$. Therefore it must be the case that $g | \gcd(m^2, n^2) = 1$ and so $g = 1$. A similar story holds for $\gcd(\frac{n^2-m^2}{2}, n)$ and so $\gcd(\frac{n^2-m^2}{2}, mn) = 1 \cdot 1 = 1$.

→ The idea is nearly identical for $\gcd(mn, \frac{n^2+m^2}{2}) = 1$.

→ For $\gcd(\frac{n^2-m^2}{2}, \frac{n^2+m^2}{2}) = g$, we have $g | \frac{n^2-m^2}{2}, \frac{n^2+m^2}{2}$ and so $g | \frac{n^2-m^2+n^2+m^2}{2} = n^2$ and $g | \frac{n^2+m^2-(n^2-m^2)}{2} = m^2$. Therefore we must have $g | \gcd(n^2, m^2) = 1$ which implies $g = 1$.

(3) Suppose there were some other pair (a, b) which gave rise to the same PPT's.

Then we would have that $(\frac{b^2-a^2}{2}, ab, \frac{b^2+a^2}{2}) = (\frac{n^2-m^2}{2}, mn, \frac{n^2+m^2}{2})$. We then NTS

$m = a$ and $n = b$. Notice $\frac{b^2+a^2}{2} - \frac{b^2-a^2}{2} = \frac{n^2+m^2}{2} - \frac{n^2-m^2}{2}$ implies $\frac{2a^2}{2} = \frac{2m^2}{2}$ so

$a^2 = m^2$. Then WLOG, we may assume a, b, m, n are positive, thus $a = m$. Lastly,

$\frac{b^2-a^2}{2} + \frac{b^2+a^2}{2} = \frac{n^2-m^2}{2} + \frac{n^2+m^2}{2}$ implies $b^2 = n^2$ and therefore $b = n$. $\square$

Suppose we wanted to find integral solutions to $x^2 + y^2 = z^2$. The solutions (x, y, z) are called Pythagorean Triples.

→ We concern ourselves with the number of solutions. In general, the number of solutions is not difficult to classify: either there is no solution or infinitely many.

⚠ Note: Strictly speaking, if we allow for $x, y \in \mathbb{Z}$ there are infinitely many Pyth. trip. of the form $(0, z, z)$. If we disregard this trivial solution, then the above holds.

→ To see why there are infinitely many Pyth. trip. consider the example:

Notice $3^2 + 4^2 = 9 + 16 = 25 = 5^2$. Therefore, ~~the~~ $(3, 4, 5)$ is a Pyth. trip.

But then since $3^2 + 4^2 = 5^2$, $4(3^2 + 4^2) = 4 \cdot 5^2$ or $(3 \cdot 2)^2 + (4 \cdot 2)^2 = (5 \cdot 2)^2$ and

So $(6, 8, 10)$ is also a Pyth. trip. We can continue to multiply by squares to see that $(3k, 4k, 5k)$ is a Pyth. trip. for $k \in \mathbb{Z}$.

→ We refer to the solution $(3k, 4k, 5k)$ as a family. We could then instead ask if there are infinite disjoint families of Pyth. trip.

→ To do so, we define a primitive Pythagorean triple to be the 3-tuple (x, y, z)

s.t. $\gcd(x, y) = \gcd(x, z) = \gcd(y, z) = 1$. That is, they're ^{pairwise} relatively prime.

Example: The 3-tuple $(3, 4, 5)$ is a primitive Pyth. trip. (PPT) because $\gcd(3, 4) = 1$, $\gcd(3, 5) = \gcd(4, 5) = 1$. However, the 3-tuple we built, $(6, 8, 10)$, is not. Since $\gcd(6, 8) = 2 = \gcd(6, 10) = \gcd(8, 10) \neq 1$. Thus the PPT can be thought of as the "canonical" representation of that family of Pyth. trip.

→ The above example can serve as a heuristic for the following.

Lemma: Every Pythagorean triple arises from a unique PPT.

Theorem: All pythagorean triples consist of disjoint infinite families which are obtained by scaling a PPT.

→ Now that we've considered PPT's our query becomes can we describe or classify all PPT's?

→ First observation: If (x, y, z) is a PPT, then z is odd.

Proof: If (x, y, z) is a PPT, then $x^2 + y^2 = z^2$ and $\gcd(x, y) = \gcd(x, z) = \gcd(y, z) = 1$. Then recall that for all $n \in \mathbb{Z}$, $n^2 \equiv 0, 1 \pmod{4}$. Therefore, if we consider $x^2 + y^2$ modulo 4, we have $x^2 + y^2 \in \{0, 1, 2\}$. Now, for the sake of contradiction, suppose z were even. If z were even, then $x^2 + y^2 = z^2 \equiv 0 \pmod{4}$. However $x^2 + y^2 \equiv 0 \pmod{4}$ implies $x^2 \equiv 0 \pmod{4}$ and $y^2 \equiv 0 \pmod{4}$, or that x^2 & y^2 are even. If x^2 & y^2 are even, then x & y are even, contradicting the gcd constraint we have. $\square$

→ Second observation: one of x & y is odd and the other is even.

Proof: This simply follows from the facts that $(\text{odd})^2 = \text{odd}$, $(\text{even})^2 = \text{even}$, $\text{even} + \text{even} = \text{even}$, $\text{odd} + \text{odd} = \text{even}$, and $\text{odd} + \text{even} = \text{odd}$. Therefore, the only possibility for $x^2 + y^2 = \text{odd}$ is for one of x & y to be odd and the other even. $\square$

→ Third observation: A formula for x, y, z .

Proof: Suppose WLOG that x is even. Then write $x^2 + y^2 = z^2$ as $y^2 = z^2 - x^2 = (z-x)(z+x)$. Let $\gcd(z-x, z+x) = g$ so that $g \mid z-x, z+x$ which implies $g \mid z-x + (z+x) = 2z$ and $g \mid z+x - (z-x) = 2x$. Moreover, with z odd and x even, $z \pm x = \text{odd}$ and so g cannot be even. Then, $g \mid x, z$ iff $g \mid \gcd(x, z) = 1$ which implies $g = 1$.

Now, if $z-x = m^2$ and $z+x = n^2$, then $y = mn$, $z = \frac{n^2 + m^2}{2}$, and $x = \frac{n^2 - m^2}{2}$.

Here m, n are odd and $\gcd(m, n) = 1$.

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Example: Suppose we wanted to find all $n \in \mathbb{Z}_{>0}$ for which $\varphi(n) = k$ for $k \in \mathbb{Z}_{>0}$.
Given $k=1$, the solution is clear: $n \in \{1, 2\}$. By definition, $\varphi(1)=1$. Then $\varphi(2)=2-1=1$.
We then consider a lemma.

Lemma: Given $p|n$, $p-1 | \varphi(n)$ for p prime.

Proof: Recall $\varphi(n) = n \cdot \prod_{p|n} (1 - \frac{1}{p})$ for distinct primes p . Then $(1 - \frac{1}{p}) = (\frac{p-1}{p})$,
and since $p|n \Rightarrow \frac{n}{p} \in \mathbb{Z}$, $\varphi(n) = n \cdot \prod_{p|n} (\frac{p-1}{p}) = (p-1) \cdot k$ for $k \in \mathbb{Z}$. This
therefore implies $p-1 | \varphi(n)$. $\square$

Corollary: For all $n > 2$, $\varphi(n) \in 2\mathbb{Z}$.

Proof: There are two cases: either n has an odd prime factor or it doesn't. Suppose
 n did contain an odd prime factor, p . Then since $(2k+1)-1 = 2k$ is even,
 $p-1$ will be even, which implies $2k | \varphi(n)$ or $\varphi(n) = 2kt$ is even. Furthermore,
we may restrict ourself to the consideration that n has one prime factor
without loss of generality since $(\text{odd})^2 = \text{odd}$ and $\text{odd}-1 = \text{even}$.

Now suppose n has no ~~prime~~ odd prime factors and so $n = 2^k$. We then
have $\varphi(2^k) = 2^k - 2^{k-1} = 2(2^{k-1} - 2^{k-2})$, which is even. $\square$

Example Continued:

By the above work, if $k=1$, then $n \in \{1, 2\}$. If k is odd but greater than
one, then no such n exists. Otherwise, if k is even, list all of its divisors. Then
determine which ones have $p-1 = d$ true where p is prime and d is a divisor. These
will form the candidate solutions for $n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$ where the $p_i^{e_i}$ are from
above. Then start systematically determining the n for which $\varphi(n) = k$ is true
by starting with the largest prime p^* and letting $n = p^* \cdot q_1$ where $\gcd(p^*, q_1) = 1$
and q_1 is any multiple of the p_i 's.

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Example: Find all $m, n \in \mathbb{Z}$ such that $2mn + m - 2n = 7$.

Like the ~~totient~~ inverse totient problem, our search space for all possible m, n is infinite. Thus we need to reduce the problem to a finite one somehow. We do so by noticing $(2n+1)(m-1) = 2nm - 2n + m - 1$. Therefore we have that $(2n+1)(m-1) = 2nm + m - 2n - 1 = 7 - 1 = 6$. The divisors of 6 in $\mathbb{Z}$ are $-1, -2, -3, -6, 6, 3, 2, 1$ and since $2n+1$ is always odd, it can only be $\pm 1, \pm 3$. Therefore we've reduced our infinite search to only 4 possible cases.

$$\begin{array}{llll} 2n+1=1 \Rightarrow n=0 & 2n+1=-1 \Rightarrow n=-1 & 2n+1=3 \Rightarrow n=1 & 2n+1=-3 \Rightarrow n=-2 \\ m-1=6 \Rightarrow m=7 & m-1=-6 \Rightarrow m=-5 & m-1=2 \Rightarrow m=3 & m-1=-2 \Rightarrow m=-1 \end{array}$$

So the only solutions (m, n) are $(7, 0)$, $(-5, -1)$, $(3, 1)$, and $(-1, -2)$.

Theorem: If p is prime, then p is the sum of two squares iff $p=2$ or $p \equiv 1 \pmod{4}$.

Proof: We will only prove the implication right now. We prove the converse next lecture.

We NTS if p is the sum of two squares then $p=2$ or $p \equiv 1 \pmod{4}$.

Thus suppose $p = x^2 + y^2$ for $x, y \in \mathbb{Z}$. Then notice that if $x \in \mathbb{Z}/4\mathbb{Z}$, then $x^2 \in \{0, 1\} \subset \mathbb{Z}/4\mathbb{Z}$. Therefore, $x^2 + y^2 \in \{0, 1, 2\} \subset \mathbb{Z}/4\mathbb{Z}$. So we have $p \equiv 0, 1, 2 \pmod{4}$. If $p \equiv 0 \pmod{4}$, then $p = 4k$ for $k \in \mathbb{Z}$ which contradicts p being prime. If $p \equiv 1 \pmod{4}$, then we have no contradiction. Lastly if $p \equiv 2 \pmod{4}$, then $p-2 = 4k$ or $p = 2(2k+1)$. If $k \neq 0$, then we have a contradiction, so $p=2$ instead of $p \equiv 2 \pmod{4}$. $\square$

Math 115 Lecture 11 July 8, 2026

Cyclotomic polynomial Theorem:

Over any field F , for all $d \in \mathbb{Z}_{>0}$, there is some polynomial $\Phi_d(x) \in F[x]$ for which:

(1) $a \in F^\times$ has order d iff a is a root of $\Phi_d(x)$

(2) $\Phi_d(x)$ has degree $\varphi(d)$

(3) For all $n \in \mathbb{Z}_{>0}$, $x^n - 1 = \prod_{d|n} \Phi_d(x)$

(4) $\Phi_n(x) = \prod_{d|n} (x^d - 1)^{\mu(n/d)}$

A Recursive Formula:

If we use (3) from above and write $\prod_{d|n} \Phi_d(x) = \prod_{d|n, d \neq n} \Phi_d(x) \Phi_n(x)$, we get the recursive formula $\Phi_n(x) = \frac{x^n - 1}{\prod_{d|n, d \neq n} \Phi_d(x)}$.

n -th roots of unity:

We denote the n th roots of unity by $\zeta_n = e^{2\pi i/n}$. These are the solutions to $x^n - 1 = 0$ over $\mathbb{C}$.

Primitive ~~nth~~ n th root of unity:

We say ζ_n^k is primitive iff $\gcd(n, k) = 1$. These will generate every n th root of unity.

Since there are $\varphi(n)$ values for which $\gcd(n, k) = 1$, there are $\varphi(n)$ primitive n th roots of unity. Moreover, if $\gcd(n, k) > 1$, then $(\zeta_n^k)^{n/k} = 1$. Lastly, these are the roots of $\Phi_n(x)$ over $\mathbb{C}$.

Note: (Only over $\mathbb{C}$) We define $\Phi_n(x) = \prod_{\substack{1 \leq k \leq n \\ \gcd(n, k) = 1}} (x - e^{2\pi i k/n})$

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→ Euler Totient: Recall for all $n \in \mathbb{Z}_{>0}$, $n = \prod_{i=1}^k p_i^{e_i}$, $p_i \neq p_j$ prime, $\varphi(n) = n \cdot \prod_{i=1}^k (1 - \frac{1}{p_i})$.

↳ Examples: (1) If p is prime, $\varphi(p) = p(1 - \frac{1}{p}) = p-1$

(2) If p is prime, $\varphi(p^k) = p^k(1 - \frac{1}{p}) = p^{k-1}(p-1) = p^k - p^{k-1}$.

(3) $\varphi(31) = 30$, $\varphi(9) = \varphi(3^2) = 3^2 - 3 = 6$, $\varphi(1000) = \varphi(2^3 \cdot 5^3) = \varphi(2^3) \varphi(5^3) = 400$

Theorem: For $F: \mathbb{Z}_{>0} \rightarrow \mathbb{C}$ multiplicative, $G(n) = \sum_{d|n} F(d)$ is also multiplicative.

proof: We NTS $G(mn) = G(m) \cdot G(n)$. Note $G(mn) = \sum_{d|mn} F(d)$. Then, if $\gcd(m, n) = 1$, there exists unique d_1, d_2 for which $d = d_1 d_2$, $d_1 | m$, $d_2 | n$, and $\gcd(d_1, d_2) = 1$.

Therefore write $G(mn) = \sum_{d|mn} F(d) = \sum_{d_1|m, d_2|n} F(d_1 d_2) = \sum_{d_1|m, d_2|n} F(d_1) F(d_2) = \underbrace{\left(\sum_{d_1|m} F(d_1) \right)}_{G(m)} \cdot \underbrace{\left(\sum_{d_2|n} F(d_2) \right)}_{G(n)}$

Which completes the proof. $\square$

Example: For $\alpha \in \mathbb{R}$, the function $n \mapsto n^\alpha$ is totally multiplicative. To see this, notice that $mn \mapsto (mn)^\alpha = m^\alpha \cdot n^\alpha$ by the law of exponents. Moreover, by our theorem, the function $\sigma_\alpha(n) = \sum_{d|n} d^\alpha$ is also multiplicative.

Example: Let $n = 3$, the divisors of 3 are 1, 3. The divisors of 7 are 1, 7. The divisors of 21 are 1, 3, 7, 21. Set $\alpha = 2$, then from above, we expect $\sigma_2(21) = \sigma_2(3) \cdot \sigma_2(7)$. We have $\sigma_2(21) = 1^2 + 3^2 + 7^2 + 21^2$, $\sigma_2(3) = 1^2 + 3^2$, $\sigma_2(7) = 1^2 + 7^2$, and $\sigma_2(3) \sigma_2(7) = (1 + 3^2)(1 + 7^2) = 1 + 3^2 + 7^2 + 3^2 \cdot 7^2 = 1 + 3^2 + 7^2 + 21^2 = \sigma_2(21)$.

Note: We can write $\sigma_\alpha(n)$ in closed form.

Write $n = \prod_{i=1}^k p_i^{e_i}$ so $\sigma_\alpha(n) = \sigma_\alpha(p_1^{e_1}) \sigma_\alpha(p_2^{e_2}) \cdots \sigma_\alpha(p_k^{e_k})$. Then, the divisors of $p_i^{e_i}$ are $1, p_i, p_i^2, \dots, p_i^{e_i-1}, p_i^{e_i}$, in which $\sigma_\alpha(p_i^{e_i}) = 1^\alpha + p_i^\alpha + p_i^{2\alpha} + \dots + p_i^{e_i\alpha} = \frac{p_i^{(e_i+1)\alpha} - 1}{p_i^\alpha - 1}$

by the finite geometric sum formula. Therefore,

$$\sigma_\alpha(n) = \prod_{i=1}^k \left(\frac{p_i^{(e_i+1)\alpha} - 1}{p_i^\alpha - 1} \right).$$

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Theorem: Two multiplicative functions F and G are equal iff $F(p^n) = G(p^n)$ for all primes.

Proof: First, for $n=0$, $F(1) = 1 = G(1)$, so suppose $n > 0$. First, suppose $F = G$ as multiplicative functions. Since $F, G: \mathbb{Z}_{>0} \rightarrow \mathbb{C}$ and $p^n \in \mathbb{Z}_{>0}$, we necessarily have $F(p^n) = G(p^n)$.

Conversely, suppose $F(p^n) = G(p^n)$ and we ~~must~~ NTS $F = G$. Since $\mathbb{Z}$ is a UFD, write $n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$ so $F(n) = F(p_1^{e_1}) F(p_2^{e_2}) \dots F(p_k^{e_k})$
 $= G(p_1^{e_1}) G(p_2^{e_2}) \dots G(p_k^{e_k}) = G(n)$,

completing the proof. $\square$

Theorem: For all $n > 0$, $\sum_{d|n} \varphi(d) = n$.

Proof: First, φ is clearly multiplicative. Moreover, with $\varphi(n)$ multiplicative, by our previous theorem, $I(n) := \sum_{d|n} \varphi(d)$ is multiplicative. Therefore, by the above theorem, we have equality iff $I(p^k) = p^k$. We have $I(p^k) = \sum_{d|p^k} \varphi(d)$ and the numbers which divide p^k are $1, p, p^2, \dots, p^k$, so $I(p^k) = \sum_{i=0}^k \varphi(p^i)$, but then recall $\varphi(p^i) = p^i - p^{i-1}$. Therefore, $I(p^k) = 1 + (p-1) + (p^2-p) + (p^3-p^2) + \dots + (p^{k-1}-p^{k-2}) + (p^k-p^{k-1})$ and so $I(p^k) = p^k$. $\square$

Theorem: The Möbius function μ is multiplicative but not totally multiplicative.

Proof: Recall if $n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$, then $\mu(n) = (-1)^k$. Thus if $m = q_1 q_2 \dots q_j$, then $mn = p_1 p_2 \dots p_k q_1 q_2 \dots q_j$ in which $\mu(mn) = (-1)^{k+j} = (-1)^k (-1)^j = \mu(n) \mu(m)$ and so μ is multiplicative.

To see μ is not totally multiplicative notice $\mu(4) = 0$ and $\mu(2) = -1$. Thus $\mu(2)^2 = (-1)^2 = 1 \neq 0 = \mu(4)$. $\square$

Theorem: $\sum_{d|n} \mu(d)$ is a multiplicative function, $\sum_{d|1} \mu(d) = 1$, and $\sum_{d|n} \mu(d) = 0$ for $n > 1$.

Proof: First, $\sum_{d|n} \mu(d)$ is a multiplicative function since μ is. Then the only divisor of 1 is 1 and so $\sum_{d|1} \mu(d) = \mu(1) = 1$. If $n > 1$, then write $n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$. Then $\sum_{d|p^k} \mu(d) = \mu(1) + \mu(p) + \mu(p^2) + \dots + \mu(p^k) = 1 - 1 = 0$. We then have $\sum_{d|n} \mu(d) = \prod_{i=1}^k \left(\sum_{d|p_i^{e_i}} \mu(d) \right) = \prod_{i=1}^k 0 = 0$. $\square$

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Möbius Inversion Theorem: Let $F: \mathbb{Z}_{>0} \rightarrow \mathbb{C}$ be any function (not necessarily multiplicative).

Then if $G(n) = \sum_{d|n} F(d)$, we have $F(n) = \sum_{d|n} \mu(d) \cdot G(\frac{n}{d})$.

Proof: We have $G(\frac{n}{d}) = \sum_{d_1 | \frac{n}{d}} F(d_1)$ and so $\sum_{d|n} \mu(d) G(\frac{n}{d}) = \sum_{d|n} \mu(d) \sum_{d_1 | \frac{n}{d}} F(d_1)$. Then $d_1 | \frac{n}{d}$ implies $d_1 d | n$ and so $d_1 | n$. Then

$$\sum_{d|n} \mu(d) G(\frac{n}{d}) = \sum_{\substack{d_1 | n \\ d, d_1 | n \\ d, d_1 | n}} \mu(d) F(d_1) \quad \text{and} \quad \sum_{d|n} \mu(d) = \begin{cases} 1, n=1 \\ 0, n>1 \end{cases}. \quad \text{Thus } \cancel{\sum_{d|n} \mu(d) G(\frac{n}{d}) = F(n)}$$

$$\sum_{d|n} \mu(d) G(\frac{n}{d}) = \sum_{d_1 | n} \left(\sum_{d, d_1 | n} \mu(d) F(d_1) \right) = \mu(1) F(n) = F(n) \quad \square$$

Examples:

We have $G(1) = \sum_{d|1} F(d) = F(1)$, $G(2) = F(1) + F(2)$, $G(3) = F(1) + F(3)$, $G(4) = F(1) + F(2) + F(4)$.

Then $F(2) = G(2) - F(1) = G(2) - G(1)$, $F(3) = G(3) - F(1) = G(3) - G(1)$,

$F(4) = G(4) - F(1) - F(2) = G(4) - G(1) - G(2) + F(1) = G(4) - G(2)$.

Lastly, using our formula, $F(4) = \mu(1)G(4) + \mu(2)G(2) + \mu(4)G(1) = G(4) - G(2)$.